

Diffusion-Driven Self-Supervised Learning for Shape Reconstruction and Pose Estimation

Jingtao Sun, Yaonan Wang, Mingtao Feng, Chao Ding, Mike Zheng Shou and Ajmal Mian

Abstract—Fully-supervised category-level pose estimation aims to determine the 6-DoF poses of unseen instances from known categories, requiring expensive manual labeling costs. Recently, various self-supervised category-level pose estimation methods have been proposed to reduce the requirement of the annotated datasets. However, most methods rely on synthetic data or 3D CAD model, and they are typically limited to addressing single-object pose problems without considering multi-objective tasks or shape reconstruction. To overcome these challenges and limitations, we introduce a diffusion-driven self-supervised network for multi-object shape reconstruction and categorical pose estimation, only leveraging the shape priors. Specifically, to capture the SE(3)-equivariant pose features and 3D scale-invariant shape information, we present a Prior-Aware Pyramid 3D Point Transformer. This module adopts a point convolutional layer with radial-kernels for pose-aware learning and a 3D scale-invariant graph convolution layer for object-level shape representation. Furthermore, we introduce a Pretrain-to-Refine Self-Supervised Training Paradigm to train our network. It enables proposed network to capture the associations between shape priors and observations, addressing the challenge of intra-class shape variations by utilising the diffusion mechanism. Extensive experiments conducted on four public datasets and a self-built dataset demonstrate that our method significantly outperforms state-of-the-art self-supervised category-level baselines and even surpasses some fully-supervised instance-level and category-level methods. The project page is released at Self-SRPE.

Index Terms—Category-Level Pose Estimation, Shape Reconstruction, 3D Transformer, Diffusion Model, Self-Supervised Learning.



1 INTRODUCTION

OBJECT 6-DoF (Degrees-of-Freedom) pose estimation is a long-standing problem in computer vision and real-world applications, such as robotics, autonomous driving and virtual reality. Currently, most existing works focus on instance-level methods, which rely heavily on expensive manual pose labels and precise 3D CAD models. However, collecting ground-truth pose labels and models is typically time-consuming and labor-intensive. In an effort to alleviate the need for extensive labeling, recent research has explored self-supervised [1], [2], [3], [4], [5], semi-supervised [6] and weakly-supervised [7] methods for pose estimation. Nevertheless, many of these approaches still rely heavily on 3D models and perform suboptimally compared to fully-supervised methods due to the lack of manual pose label.

To reduce the dependency on object's 3D CAD model, recent researches have shifted towards the category-level pose estimation. The goal of this task is to estimate size and pose for unseen objects within the same category and the intra-class variations between object shape and texture makes this type of problem more challenging. The pioneering work is NOCS [8], as depicted in Fig. 1 (a), which utilizes implicit coordinate space to map all instances into the unified representation. Despite achieving good results,

existing category-level methods such as [9], [10], [11], typically require fully-supervised training with detailed 6-DoF pose annotations. To address this limitation, recent works [12], [13], [14], [15], [16], [17], [18] have leveraged the self-supervision training with unlabeled synthetic and real-world data for categorical pose estimation. In [12], [13], the synthetic data was used to pre-train a deformation network or transferred to real domain using pseudo labels. In [14], 3D CAD models from synthetic data are employed to learn categorical shape information. However, the synthetic-to-real (Sim2Real) domain gap often renders these methods unfeasible and impractical for real-world applications. Alternative available approaches either utilize equivariant shape analysis [15], [16] or 2D-3D correspondence [17], [18].

Nevertheless, these approaches require strong canonical meshes during the training stage. Most state-of-the-art methods for self-supervised category-level 6-DoF pose estimation focus solely on single objects, rendering them unsuitable for multi-object pose estimation. Moreover, these methods only tackle the pose estimation problem and neglect the corresponding shape reconstruction.

In this paper, we extend conventional single-object pose estimation to encompass multi-object shape reconstruction and categorical pose estimation tasks under a self-supervised manner. Our proposed tasks aim to estimate 6-DoF poses and 3D shapes of multiple surrounding instances in the observed scene. The main challenges of our proposed tasks are threefold: (i) addressing the fundamental issue of intra-class shape variations for multiple instances, which remains unresolved; (ii) reconstructing 3D shapes for different instances within the same category; (iii) estimating pose and 3D scale without ground-truth pose labels, 3D object's CAD models, or even the Sim2Real setting.

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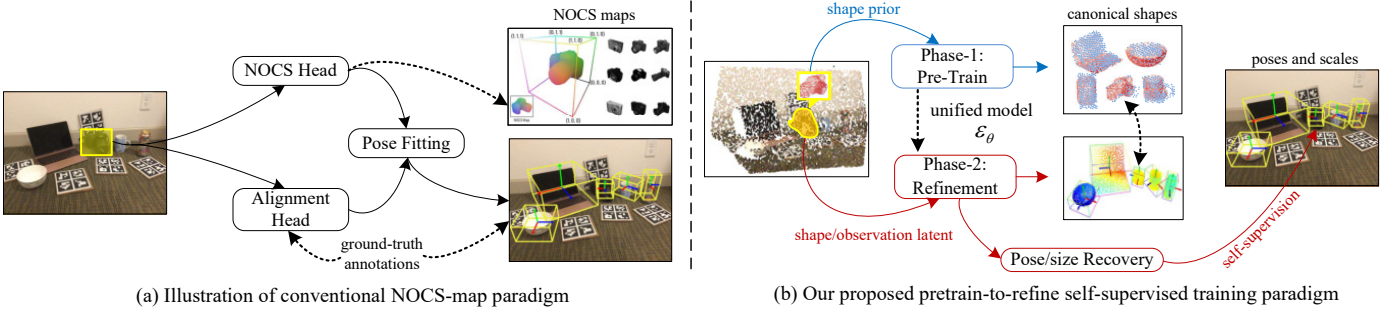


Fig. 1. Overview. Different from most existing NOCS-map paradigm, that employs the one-shot pipeline to normalize objects into a 3D NOCS space contained within a unit cube and aligns their centers or orientations within the same category, our proposed pretrain-to-refine paradigm adopts a two-phase strategy. We first train a basic network model using the prior shapes, and then fine-tune this pre-trained base model under the guidance of the shape/observation latent representations. The entire process is implemented in a self-supervised manner driven by the diffusion mechanism.

To tackle the above-mentioned challenges, we introduce a diffusion-driven self-supervised network for multi-object shape reconstruction and category-level pose estimation tasks. The key challenge of these tasks lies in capturing SE(3)-equivariant pose features and 3D scale-invariant shape information. To this end, we introduce the Prior-Aware Pyramid 3D Point Transformer module as core structure of our network. Especially, we propose a point convolutional layer with radial-kernels to densely extract category-level SE(3) equivariant pose-aware features for pose estimation, and present a 3D scale-invariant graph convolution layer to learn object-level 3D shape scale-invariant features for shape reconstruction, respectively. Furthermore, as shown in Fig. 1 (b), to implement self-supervised learning of our proposed network, we introduce a Pretrain-to-Refine Self-Supervised Training paradigm, including two phases. In the pre-training phase, we employ the inverse diffusion mechanism to enable our network to learn the priori knowledge of shape and establish the pre-trained base model. In the refinement phase, we then fine-tune this pre-trained model through self-supervised training by conditioning shape/observation latent embeddings of the object-aware 3D surface information. On top of that, our final refined model captures the shape association between shape priors and current observations to cope with the challenge of intra-class shape variations for every instance. Experiments on four widely-used benchmark datasets (REAL275 [8], CAMERA25 [8], Wild6D [19] and YCB-Video [20]), and our self-built testset DYNAMIC45 demonstrate that our approach can exhibit state-of-the-art performances. We summarize our contributions as follows:

- We present a self-supervised network for multi-object shape reconstruction and category-level 6-DoF pose estimation tasks utilizing diffusion mechanism, in which only the shape priors are required.
- We introduce the Prior-Aware Pyramid 3D Point Transformer module to explore both the SE(3)-equivariant 6-DoF pose features and 3D scale-invariant shape information, effectively capturing the invariant/equivariant property for each object.
- We propose the Pretrain-to-Refine Self-Supervised Training paradigm, designed to train our network in a self-supervised manner, while effectively solving the challenge of intra-class shape variations.

- Extensive experiments on four public datasets and our data show that our proposed method achieves state-of-the-art performance for self-supervised category-level pose estimation and comparable results with some category-level fully-supervised approaches. Additionally, our model outperforms some instance-level self/fully supervised baselines.

2 RELATED WORK

2.1 Self-Supervised Pose Estimation

Self-supervised learning has found applications in various domains such as 2D image/video understanding, depth estimation and autonomous driving, etc. Several recent works [1], [2], [3], [4], [5], [6], [7], [21], [22], [23] have investigated self-supervised learning for instance-level pose estimation. In [21], Deng *et al.* proposed a self-labeling pose estimation method for robotic manipulation. In [22], Sock *et al.* leveraged the visual consistency to refine pose estimation under the supervision of ground-truth masks. Self6D [1] and Self6D++ [2] were introduced for monocular pose estimation using a synthetic-to-real formulation. In contrast to these common render-and-compare pipelines, TexPose [3] decomposed self-supervision for 6D object pose into texture learning and pose learning, while Hai *et al.* [4] proposed a geometry-guided learning framework without any annotations. Some works aim to cover object's pose through weakly-supervised [7] or semi-supervised manners [6]. However, these instance-level self-supervised 6D object pose estimation methods can only work with additional depth information or rely on accurate annotation of 2D segmentation masks or object 3D meshes, limiting their application range.

In contrast, we aim to explore self-supervised category-level pose estimation in this paper. To date, few works have concentrated on self-supervised category-level pose estimation, including [12], [13], [14], [15], [16], [17], [18], [24], [25], [26]. In [16], Li *et al.* proposed a method that leverages SE(3) equivalent representation for category-level pose estimation, while Liu *et al.* [15] presented a part-level SE(3) equivariance for articulated object pose estimation. CPS++ [25] optimized pose results utilizing the consistency between the observed depth map and the rendered depth. Recently, the unsupervised domain adaptation schemes

have been used for this task, such as UDA-COPE [12] and Self-DPDN [13]. SSC-6D [14] leveraged DeepSDF as a 3D representation to improve the pose estimation performance. Additionally, Zaccaria *et al.* [17] utilized 2D optical flow as a proxy for supervising 6D pose.

Our proposed network is also a self-supervised method for category-level 6-DoF pose estimation. However, unlike existing self-supervised category-level approaches that leverage the synthetic data/model, depth image or pseudo labels as additional supervision signals, we only require shape priors to implement self-supervised learning by capturing the association between prior knowledge and observations. This effectively avoids the challenge of sim2real gap. Furthermore, our method not only achieves self-supervised pose estimation but also accomplishes the task of shape reconstruction for multiple objects.

2.2 Fully-Supervised Pose Estimation

Previous studies in pose estimation have primarily focused on the problem at the *instance-level* [2], [27], [28], [29], [30], [31], [32], [33], [34], [35]. Few works have specifically addressed *category-level* pose estimation, and the majority of these approaches employ fully supervised learning [36], [37], [38], [39], [40], [41], [42]. Wang *et al.* [8] introduced the pioneering work called NOCS, which handles various unseen objects within a category and estimates the 6D pose and dimensions by inferring the correspondence between the observation and the shared NOCS. According to NOCS, Chen *et al.* [43] introduced CASS, a unified representation that models the latent space of canonical shape with normalized pose. Additionally, SPD [10] and SGPA [9] employed the shape prior of the observed object to obtain the deformation matrix and structure similarity. Zou *et al.* introduced 6D-ViT in [44], a Transformer-based method that captures the correspondence matrix and deformation field. CenterSnap [45] introduced an approach that simultaneously predicts the 3D shape and estimates the 6D pose and size without relying on bounding boxes. However, all of these methods employ fully-supervised training, relying on ground-truth pose annotations and explicit CAD models during training stage. Different from these available methods, we address category-level pose estimation problem in a self-supervised manner. Additionally, in contrast to previous works, such as [9], [10], [45], we tackle the challenge of intra-class variations by utilizing the diffusion mechanism to capture the shape association between the observations and priors.

2.3 3D Shape Reconstruction and Completion

Shape reconstruction and completion present challenges related to generating and estimating objects in 3D space. Initially, researchers attempted to adapt 2D methods to 3D point clouds through techniques such as 3D convolution or voxelization, or by directly processing 3D coordinates. These efforts were inspired by the remarkable performance achieved by PointNet [46] and PointNet++ [47]. To date, existing approaches can be categorized into point-based (e.g., PCN [48], FinerPCN [49], SK-PCN [50], MSPCN [51], and SRPCN [52]), view-based [53], [54], convolution-based, graph-based [55], transformer-based [56], [57],

TABLE 1
Concise notation of symbols. The main symbols are clarified.

Symbol	Description
$P_0 \in \mathbb{R}^{N_0 \times 9}$	Observable point cloud
$P_r \in \mathbb{R}^{N_r \times 3}$	Corresponding point cloud of shape prior
$P \in \mathbb{R}^{K \times M \times 3}$	Reconstructed canonical shape point sets
$\tilde{P}_k = \{\tilde{R}_k, \tilde{t}_k\}$	Estimated 6-DoF poses for all objects
ε_θ	Learnable parameters of proposed network
$\tilde{s}_k \in \mathbb{R}^3$	Estimated 3D scales for all objects
$F_l, F^{(l)} (l = 0, 1, 2, 3)$	Feature map and group map of each resolution
$N_x^{(i)} (i = 1, 2, 3, 4)$	The neighbor domain space
$\tilde{x}_k$	The coordinates of each radial kernel points
$x_i^t, t \in (0 \cdots T)$	Sampled noise points during diffusion process
f	The shape/observation latent embedding
$ G_g $	The scale of final pose/size hypothesis

and generative model-based methods (e.g., GAN-based approach [58] and VAE-based approach [59], [60]). What distinguishes our proposed method from these types of 3D shape reconstruction is that we aim to reconstruct the canonical 3D shapes for different instances within the same category, only leveraging the shape priors. While many existing works concentrate on single-instance shape reconstruction with supervision from specific complete point clouds, our method targets multiple objects in current observation and operates in a self-supervised manner without requiring ground-truth point clouds or models.

3 METHOD

3.1 Preliminary

Problem Statement and Notations. The goal of this work is to address the simultaneous tasks of self-supervised 3D shape reconstruction and 6-DoF pose localization for all unseen object instances in 3D space. To clearly demonstrate superior potential of our method, we focus on the stricter self-supervised setting, *w.r.t.* we trained our model without manual pose annotations, synthetic data conversions and ground-truth CAD object's models. Concretely, given the currently observable partial point cloud $P_0 \in \mathbb{R}^{N_0 \times 9}$ ($'g'$ contains coordinates, normal vector and color information) and the corresponding shape prior $P_r \in \mathbb{R}^{N_r \times 3}$ for each instance of same category, our objective is to train an end-to-end model $\Gamma_{\varepsilon_\theta}$ that outputs canonical 3D shape point sets $P \in \mathbb{R}^{K \times M \times 3}$ along with the 6-DoF poses $\tilde{P}_k = \{\tilde{R}_k \in SE(3), \tilde{t}_k \in \mathbb{R}^3\}$, and 3D scales $\tilde{s}_k \in \mathbb{R}^3$ for all currently visible instances:

$$(P, \tilde{P}_k, \tilde{s}_k) = \Gamma_{\varepsilon_\theta}(P_0, P_r), \quad (1)$$

where $k = 1, 2, \dots, K$, K is the number of arbitrary objects in the current scene and M denotes the number of points in reconstructed shapes. $\tilde{R}_k$ and $\tilde{t}_k$ are the rotation matrix and translation vector, respectively. ε_θ stands for learnable model parameters. To identify unique shape codes for each shape prior, we employ the Shape Auto-Encoder proposed in [45]. For the reader's convenience, we clarify the main mathematical symbols in our paper, as provided in Table 1.

Network Architecture. As described in Fig. 2, our network comprises three submodules: (1) Prior-Aware Pyramid 3D Point Transformer serving as the base module for latent

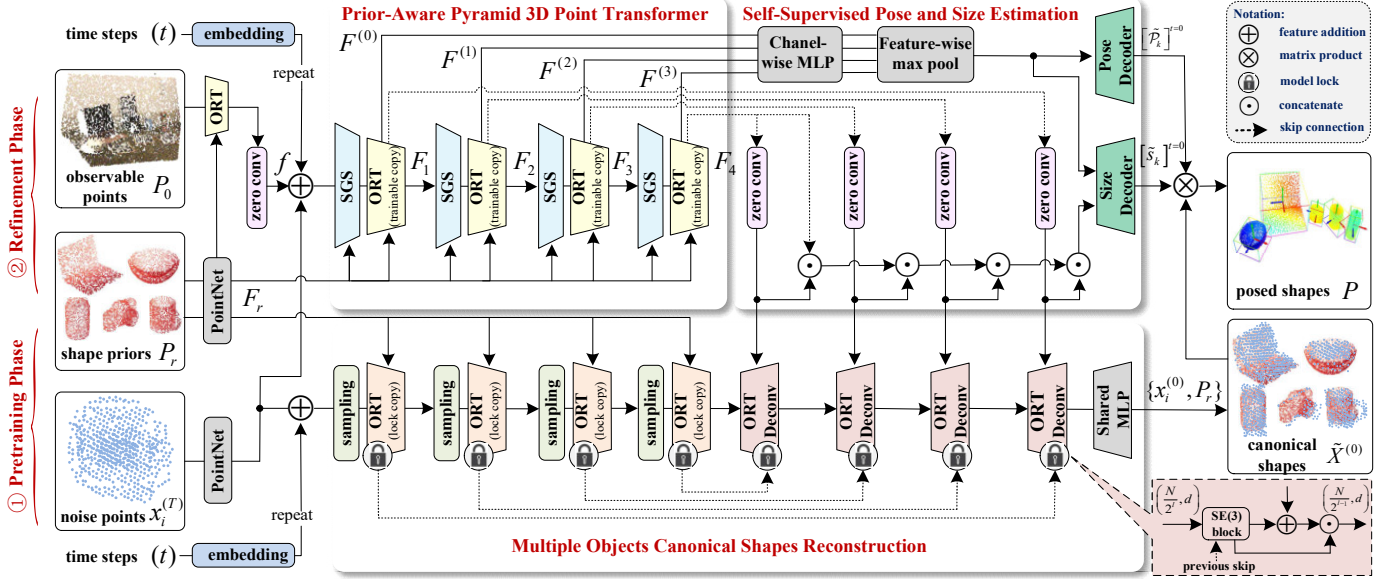


Fig. 2. Illustration of our network and pretrain-to-refine self-supervised training paradigm. Our network consists of three components, taking only the observable point cloud P_0 in the current frame and the corresponding shape priors P_r as input. Prior-Aware Pyramid 3D Point Transformer as the core network framework for self-supervised learning. During the pre-training phase, we first utilize a simplified version of our Prior-Aware Pyramid 3D Point Transformer with our ORT Deconv to establish a rough base pre-trained model, leveraging the guidance of prior shapes. After that, utilizing the shape/observation latent embedding f , we fine-tune this pre-trained model to conduct a comprehensive reinforced model in the subsequent refinement phase. Ultimately, the reinforced model is employed to determine the 6-DoF poses, 3D scales and finer canonical shapes.

representation learning between prior points and current observable point cloud (Sec. 3.2); (2) Canonical Shapes Reconstruction module for the coarse-to-fine reconstruction of 3D shapes (Sec. 3.3); and (3) Self-Supervised Pose and Size Estimation for final estimation of 6-DoF poses and 3D scales (Sec. 3.5). The following subsections will provide the detailed elaboration of each submodule, demonstrating how our pretrain-to-refine self-supervised training is implemented (refer to Sec. 3.4).

3.2 Prior-Aware Pyramid 3D Point Transformer

This subsection introduces our proposed Prior-Aware Pyramid 3D Point Transformer, taking P_0 and P_r as inputs for the following discussion. As shown in the top of Fig. 2, our proposed module includes four transformer encoding stages, generating corresponding feature maps for the observable point clouds at four resolutions. It is noteworthy that, to comprehensively model structural similarities between observable points and shape-prior points, and sufficiently convert semantic features from the prior, we have designed the same structure for all four stages, though the parameters are not shared. Given the observable points P_0 and prior points P_r , we first feed them into PointNet-based layer to extract feature embeddings $F_0 \in \mathbb{R}^{N_0 \times d}$ and $F_r \in \mathbb{R}^{N_r \times d}$, respectively. After that, we import these embeddings into the Transformer E_1 to produce the feature map F_1 and group feature map $F^{(1)}$. This process is iteratively applied through the subsequent stages E_l and E_{l+1} , generating a series of feature maps $\{F_1, F_2, F_3, F_4\}$ with size of $(N_0/2^{l+1}) \times d$ and corresponding group feature maps $F^{(l)} \in \mathbb{R}^{(N_0/2^{l+1}) \times d \times |G_l|}$ at the four resolutions ($l = 0, 1, 2, 3$). Each transformer stage E_{l+1} comprises two blocks: Shape-Guided Sampling (SGS) and Observation-Related Transformer (ORT).

Shape-Guided Sampling (SGS). Different from the conventional sampling approach in [41], [44], [61], we propose a shape-guided sampling technique to preserve more sampled points associated with the targeted instance, considering both spatial feature distance and semantic similarity between the observed point cloud P_0 and its shape prior P_r . Our key insight consists of two aspects: the region of interest of the object in the observation should exhibit a closer feature distance to prior points, and relying solely on the feature distance as the criterion may result in the majority of sampling points falling on the same instance, thus the feature semantic similarity is a nice way to avoid this matter. Specifically, given the each stage feature map F_{l+1} generated from the previous transformer module E_{l+1} and the prior point cloud feature map F_r , we initially calculate the pairwise feature distance mapping matrix $D \in \mathbb{R}^{(N_0/2^{l+1}) \times N_r}$ as follows:

$$D_{ij} = \|f_i - f_j^{(r)}\|_2, \forall f_i \in F_{l+1}, \forall f_j^{(r)} \in F_r, \quad (2)$$

where $\|\cdot\|_2$ denotes L2-Norm. Subsequently, we compute each distance by considering the overall distance between each point from the current observable point cloud and all prior points in the spatial feature space:

$$\nu_i = \sum_{j=1}^{N_r} (D_{ij}), \forall i \in \{1, 2, \dots, N_0/2^{l+1}\}. \quad (3)$$

We sample a quarter of the points based on the minimum value of ν_i , while the remaining quarter sampling points are acquired via the next semantic feature similarity.

The semantic feature similarity is computed using the attention scoring mechanism. We employ the each stage feature map F_{l+1} as the group of input vectors F_{l+1}^i ($i = 1, 2, \dots, N_0/2^{l+1}$) and take the shape prior feature F_r as

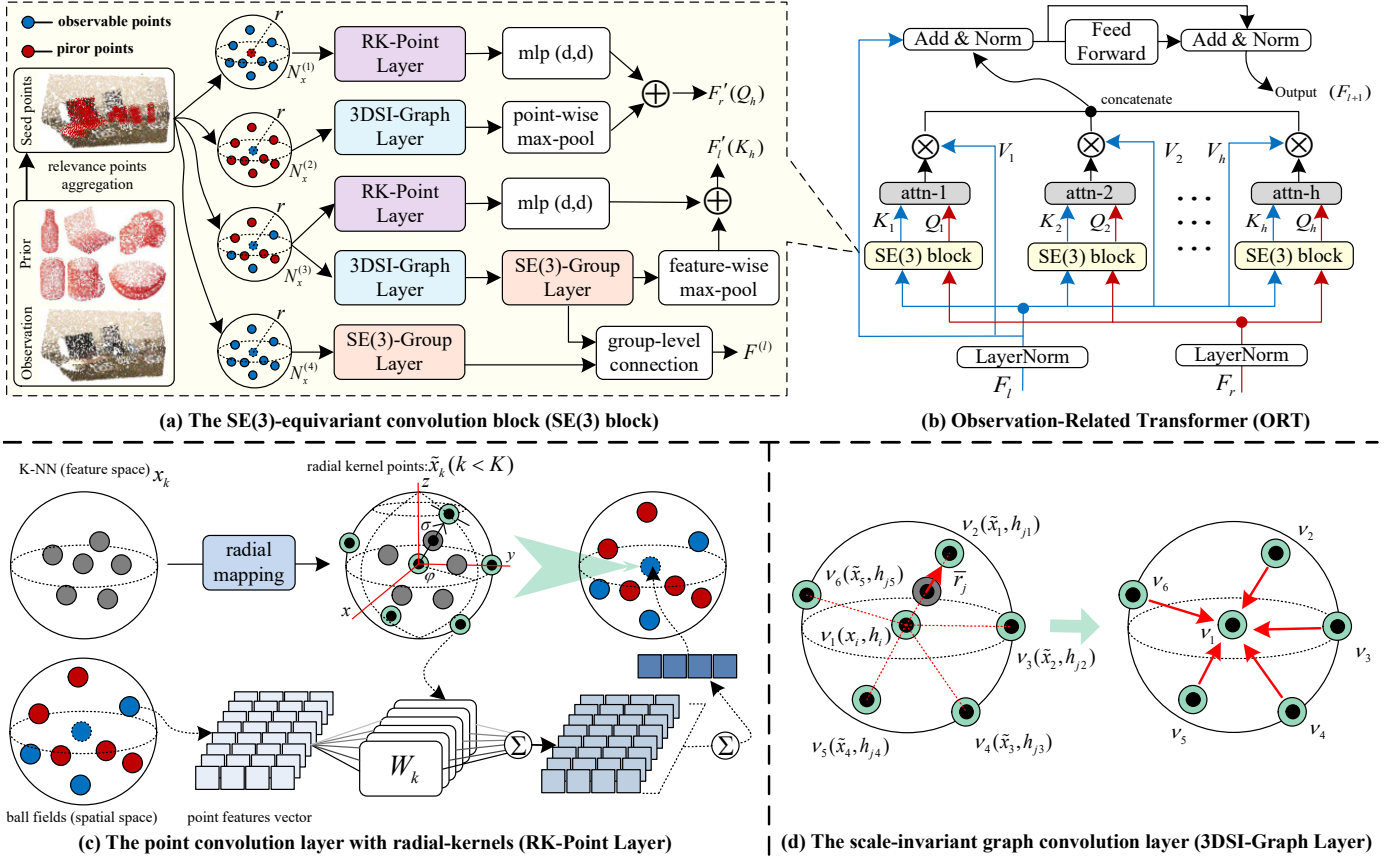


Fig. 3. Illustration of the ORT in our proposed Prior-Aware Pyramid 3D Point Transformer. (a) The proposed SE(3) block is equipped with an ability to encode shape-observation similarity based on proposed SE(3)-equivariant and 3D scale-invariant learning. (b) The details of our proposed 3D Transformer with multi-head attention, where h is the number of heads. (c) The process of the point convolution layer with radial-kernels, in which the kernel points are generated from the radial mapping on the surface of the sphere domain. (d) The 3D scale-invariant graph convolution layer, that distinguishes itself from other graph convolutions by exhibiting the scale-invariant property through the formation of a graph involving the central point and all radial kernel points.

the query vectors, respectively. The similarity score $\alpha = [\alpha_1, \alpha_2, \dots, \alpha_{N_0/2^{l+1}}]$ between F_{l+1}^i and F_r is calculated by the following formula:

$$\alpha_i = \text{softmax}(s(F_{l+1}^i, F_r)) = \frac{\exp(s(F_{l+1}^i, F_r))}{\sum_{j=1}^{N_0/2^{l+1}} \exp(s(F_{l+1}^j, F_r))}, \quad (4)$$

where $s(F_{l+1}^i, F_r) = (F_{l+1}^i)^T \cdot F_r$ is the score function. We sample the remaining quarter of points based on the maximum value of α_i .

Observation-Related Transformer (ORT). In this block, we aim to explore SE(3)-equivariant pose features and 3D scale-invariant shape information. Unlike existing works [9], [10], where the shape prior is directly used to conduct the deformation field and reconstruct the corresponding shape within the normalized space of visible objects, we leverage structure similarity and feature correlation to guide the generation of a unified seed point cloud. Then, a 3D Transformer is constructed to provide intensive characterisation for each resolution. In particular, given the observable point cloud P_l at current stage and its corresponding feature map F_l , along with the shape prior point cloud P_r and its feature map F_r , we first produce a coarse but complete seed point cloud with a size of $(N_r + \frac{N_0}{2^l}) \times 3$. This seed point cloud captures both the geometry of observations and the structure of the prior shapes. As depicted in Fig. 3 (a), to fuse the observable

points and shape priors, we first compute the feature correlation metrics for each prior point leveraging both feature maps F_r and F_l , and then select the most relevant K points in observable point set as the reference by these correlation metrics. Where the spatial center coordinate of these reference points serving as the center coordinate of these prior points. In the end, all prior points are directly embedded in observable points, constituting the seed point cloud. With the seed point cloud, we can define four neighbor domain spaces N_x based on the radius r , where the domain of definition is the ball, as shown in Fig. 3 (a). For each observable point, there are three domains, while for each prior point, there is one domain, defined as follows:

$$\begin{cases} N_x^{(1)} = \{x_i \in P_l, x \in P_r \mid \|x - x_i\| \leq r\} \\ N_x^{(2)} = \{x_i \in P_r, x \in P_l \mid \|x - x_i\| \leq r\} \\ N_x^{(3)} = \{x_i \in P_r \cup P_l, x \in P_l \mid \|x - x_i\| \leq r\} \\ N_x^{(4)} = \{x_i, x \in P_l \mid \|x - x_i\| \leq r\} \end{cases} \quad (5)$$

According to these domains, our network learns the SE(3)-equivariant and 3D scale-invariant properties between observable points and prior points by using our proposed SE(3)-equivariant convolution block (SE(3) block). As depicted in Fig. 3 (a), our SE(3) block takes P_l , P_r , F_l and F_r as inputs, yielding per-point SE(3)-equivariant feature maps F_l' and F_r' , as well as the corresponding group feature map for each level $F^{(l)}$. The SE(3) block is composed

of three key layers: 1) the point convolution layer with the radial kernels, aggregates point spatial correlation to characterise translation-equivariant properties; 2) the 3D scale-invariant graph convolution layer, captures 3D shape scale-invariance properties; 3) the SE(3) group convolution layer, computes rotation-equivariant correlation between input points and a kernel defined on the continuous SO(3) rotation group $\{G_g\}$.

1) *The Point Convolution Layer with Radial-Kernels.* Inspired by the principles of separable convolution in [62] and point convolution operation in [63], the formulation of this layer can be represented as:

$$(\Phi * h_p)(x, g) = \sum_{x_i \in N_x} \Phi(g^{-1}x_i, g)h_p(x - x_i), \quad (6)$$

where $N_x \in \{N_x^{(i)}, i = 1, 2, 3, 4\}$ and h_p is the kernel function under arbitrary rotation g defined in the neighbor domain space. Due to the kernel is defined in the mixed neighbor space $N_x^{(i)}$, rigid kernel points arranged regularly in [63] are not efficient in this situation. In this regard, we want h_p to dynamically adapt to different domain spaces. As shown in Fig 3 (c), we design a radial mapping operation to generate adaptive kernel points. Our proposed method takes similarity points set of the current point x in feature space using KNN as the initial kernel points $\{x_k | k < K\} \in N_x$, and outputs a set of K radial kernel points $\tilde{x}_k (k < K)$ for each initial kernel. The kernel function h_p is defined as:

$$h_p(x - x_i) = \sum_{k < K} \mu(x - x_i, \tilde{x}_k)W_k, \quad (7)$$

where $\{W_k | k < K\}$ are the associated weight matrices that map features from input size dimension to the output size. Similar to [63], we use the same linear correlation function, *w.r.t.* $\mu(x - x_i, \tilde{x}_k)$ as follow:

$$\mu(x - x_i, \tilde{x}_k) = \max(0, 1 - \frac{1}{\kappa} \|(x - x_i) - \tilde{x}_k\|), \quad (8)$$

where κ is the influence distance of the kernel points, and will be chosen according to the input density. We advocate this simple correlation/representation to ease gradient back-propagation when learning kernel deformations.

For radial kernel point generation, we project each initial kernel along the radius of N_x to map its corresponding radial kernel in the surface of ball field, as shown in Fig. 3 (c). It is noteworthy that we fix one of the kernel points at the center of the ball. These adaptive radial kernel points not only ensure equivalence to the translation, but also adapt to changes in 3D space of different domains. The coordinates of each radial kernel $\tilde{x}_k$ can be obtained from the following:

$$\begin{cases} \tilde{x}_k[x] = r \cdot \sin(\sigma) \cdot \cos(\varphi) \\ \tilde{x}_k[y] = r \cdot \sin(\sigma) \cdot \sin(\varphi) \\ \tilde{x}_k[z] = r \cdot \cos(\sigma) \end{cases}, \quad (9)$$

as depicted in Fig. 3(c), σ and φ are the angles of latitude and longitude of the radial kernel on the surface of ball field. These two angular parameters are calculated based on the mapping of kernel points and vary with the kernel points.

2) *The 3D Scale-Invariant Graph Convolution Layer.* To preserve the 3D scale-invariant properties of this layer, we

first construct a graph $\ell = (V, \partial)$ with the current point x_i and all radial kernel points $\tilde{x}_k$ as nodes $\nu_i \in V$, and edges information $\partial = (\alpha_{ij})$, as depicted in Fig. 3 (d). Here, x_i serves as the central point in our graph convolution, and $N(i) = \{j | (i, j) \in \partial\}$ denotes a set of neighbouring points (*radial points*). For each feature node embedding $h_i, h_j \in \mathbb{R}^{d+3}$, we consider both d-dimensional embeddings and its 3D coordinates. To obtain the feature node embeddings h_j corresponding to the radial kernel points, we map the feature embedding of the initial kernel points along the unit radial direction:

$$\begin{cases} h_j = \text{linear}(h_i^k \otimes \bar{r}_j) \\ \bar{r}_j = (x_k - \tilde{x}_k) / \|x_k - \tilde{x}_k\| \end{cases}, \quad (10)$$

then, we define our scale-invariant graph layer as follows:

$$h_{gnt} = \phi_h(h_i, \sum_{j \in N(i)} \phi_e(h_i, h_j, \alpha_{ij})), \quad (11)$$

where ϕ_h and ϕ_e denote the edge and node operations, respectively, which are commonly approximated by multilayer perceptrons. h_j^k represents the d-dimensional embeddings of initial kernels. Additionally, the mapping of feature node embeddings in Eq. (11) is only calculated in the unit radial direction and is not affected by spatial distance. Therefore, the 3D scale-invariant property can be directly observed in our proposed layer.

3) *The SE(3) Group Convolution Layer (SE(3)-Group Layer).* As shown in Fig. 3 (a), the composition of this layer is consistent with [62], aggregating information from neighboring rotation signals within the group. The icosahedron group is considered as the discrete rotation group. Ultimately, given the per-point SE(3)-equivariant feature maps F_l' and F_r' , we select to leverage the advantage of transformer network [64] to model the high-level similarity between the priori shapes and the observable objects. In details, we take F_l' , F_r' and the current stage feature map of the observable points F_l as the *query*, *key* and *value* for the following multi-head attention:

$$\mathcal{A}^{(h)} = \sigma\left(\frac{F_r' W_Q^{(h)} \cdot (F_l' W_K^{(h)})^T}{\sqrt{d}}\right) F_l W_V^{(h)}, \quad (12)$$

where, $\sigma(\cdot)$ denotes the standard softmax normalization function. $W_Q^{(h)}$, $W_K^{(h)}$ and $W_V^{(h)}$ all belong to $\mathbb{R}^{d \times d}$ and are learnable projection matrices. As depicted in Fig. 3 (b), the final output of the multi-head attention F_l is the concatenation of the h individual self-attention outputs $\mathcal{A}^{(h)}$, combined with the LayerNorm, Linear layer, ReLU and residual connections.

3.3 Multiple Objects Canonical Shape Reconstruction

In this section, the problem of shape reconstruction is treated as the reverse of a conditional diffusion generation process that is modeled as a Markov chain. Given the shape prior P_r and the shape/observation latent representation f , we can reconstruct the canonical shapes from coarse to fine in both two self-supervised training phases. We start by describing the diffusion formulation, and end with the detailed inference and implementation. For the sake of brevity in the following discussion, we assume that each sampled noise point cloud at per timestep contains a set of

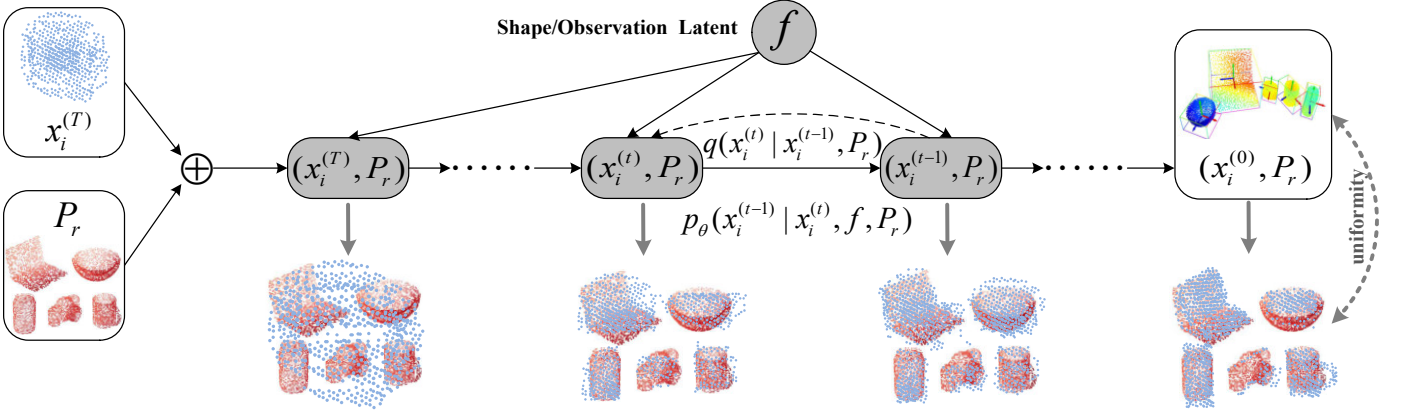


Fig. 4. The directed graphical model of diffusion process for canonical shapes reconstruction. $x_i^{(T)}$ and P_r are initial noise points and shape prior points. Each sampled noise points $x_i^{(t)}$ at per timestep contain N points. Related dynamic video can be found in the project page.

N points with xyz-coordinates, denoted as $X^{(t)} = \{\{x_i^{(t)} \in \mathbb{R}^3\}_{i=1}^N\}_{t=0}^T$.

Diffusion Formulation. As displayed in Fig. 4, similar to [65], we briefly describe examples of forward and backward processes and give each result at every process in detail. We firstly take the points from a simple noise distribution $x_i^{(T)} \sim \mathcal{N}(0, I)$ and the shape prior point clouds P_r as the inputs to construct the initial noise points $\tilde{X}^{(T)} = (x_i^{(T)}, P_r)$. These noise points are passed through the reverse Markov chain to form the complete sharp shapes regarded as the $\tilde{X}^{(0)} = (x_i^{(0)}, P_r)_{i=1}^N$. Following the conventional diffusion formulations in [65], we describe the *conditional forward diffusion process* with the number of timesteps ($t = 1, 2, 3, \dots, T$), which gradually adds the noise to $x_i^{(0)}$ while maintaining P_r unchanged. This process yields a series of shape variables with decreasing noise distribution levels, denoted as $\{(x_i^{(1)}, P_r), (x_i^{(2)}, P_r), \dots, (x_i^{(T)}, P_r)\}$. To learn our diffusion process, we define the ground-truth diffusion distribution $q(\cdot)$ by gradually adding Gaussian noise to the prior's shapes:

$$q(x_i^{(1:T)} | x_i^{(0)}) = \prod_{t=1}^T q(x_i^{(t)} | x_i^{(t-1)}, P_r), \quad (13)$$

and this Gaussian noise point provides randomness and variety to the model during the backward diffusion process. Where $q(x_i^{(t)} | x_i^{(t-1)}, P_r)$ is the Markov diffusion kernel defined as:

$$q(x_i^{(t)} | x_i^{(t-1)}, P_r) = \mathcal{N}(x_i^{(t)} | \sqrt{1 - \beta_t} x_i^{(t-1)}, \beta_t I), \quad (14)$$

with β_t representing the variance schedule parameter controlling the step size, $\{\beta_t \in (0, 1)\}_{t=1}^T$.

Since the reverse process aims to recover the expected shapes from the input noise, requiring training from data, we formulate *reverse process* as conditional denoising:

$$p_\theta(x_i^{(0:T)} | f, P_r) = p_\theta(x_i^{(T)}) \prod_{t=1}^T p_\theta(x_i^{(t-1)} | x_i^{(t)}, f, P_r), \quad (15)$$

$$p_\theta(x_i^{(t-1)} | x_i^{(t)}, f, P_r) := \mathcal{N}(x_i^{(t-1)} | \mu_\theta(x_i^{(t)}, f, P_r, t), \beta_t I), \quad (16)$$

where μ_θ represents the shape prediction parameterized by θ , and $p_\theta(x_i^{(T)})$ is a standard Gaussian prior. It is worth

highlighting that the reverse process maintains the shape prior P_r fixed and diffuses only the deformation part of inter-category shape variation between visible instances in observations and the shape priors with the same category.

Inference Details. As depicted in the bottom of Fig.2, to align with our proposed Prior-Aware Pyramid 3D Point Transformer, we adapt a unified pyramid structure for the module design of shape reconstruction task. Besides, traditional downsampling approach is directly applied without employing our SGS, owing to the input comprises $x_i^{(t)}$ and P_r . These input points originally constitute the seed point cloud, lacking practical significance in terms of noise point features. We also add the temporal embeddings into our network and use a sinusoidal position embedding along with a MLP to obtain these temporal embeddings, that contain sine and cosine pairs with varying frequencies:

$$mlp(\sin(\varpi_1 t), \cos(\varpi_1 t), \dots, \sin(\varpi_{d/2} t), \cos(\varpi_{d/2} t)), \quad (17)$$

where $\varpi_i = 1/(10^{i/d})$. Meanwhile, t and d indicate the timestep prompt and the feature dimension, respectively.

During the pretrain phase, with the prior shapes P_r , rough shapes can be reconstructed through four ORT Deconvolution (ORT-Deconv) blocks along with a shared-MLP. In our ORT-Deconv, we use the simple SE(3) block without the branch of group features extraction as the base submodule. Output features from the downsampled point set are mapped to a higher-resolution set via concatenation with the features from skip connections, as displayed in the bottom right of Fig.2. During the refinement phase, the shape/observation latent representation f is added into the trainable copy of four Transformer block in Prior-Aware Pyramid 3D Point Transformer, using a 1×1 zero convolution layer initialized with both weight and bias set to zeros. Their outputs are added to four skip-connections and then passed through four zero convolution layers, which are subsequently connected with the corresponding ORT-Deconv. Where zero convolution grows incrementally from zero to the optimized parameter in a learned manner, and is robust to our proposed self-supervised two stage training paradigm. The refiner canonical shapes can be obtained using the final refined model. Detailed inference for shape reconstruction can be found in Alg. 2.

3.4 Pretrain-to-Refine Self-Supervised Training

To enable the self-supervised training of our proposed network, we introduce a *Pretrain-to-Refine Self-Supervised Training* paradigm depicted in Fig. 2, inspired by ControlNet [66]. At the pre-training phase, we consider the canonical 3D shapes reconstruction task as a conditional diffusion generation problem where the shape priors P_r serve as the basic conditioner. We build a base model with the similar network structure of Prior-Aware Pyramid 3D Point Transformer (the bottom left of Fig. 2). At the reinforcement phase, we lock all parameters ε_θ of this model and clone it into a trainable copy, namely, per-block of Prior-Aware Pyramid 3D Point Transformer. We add the observable/shape latent embedding f as a task-specific extra condition to fine-tune the base model. Further detailed of this process can be found in Alg. 1.

Training Objective. Having formulated the conditional forward and reverse diffusion processes in Eq. (14) and (16), we now formalize the self-supervised training objective as follows. Similarly to DDPM [65], we also use the variational lower bound to optimize the negative log-likelihood of the point belonging to desired shape $\mathbb{E}_q \log p_\theta(x_i^{(0)})$:

$$-\mathbb{E}_q \log p_\theta(x_i^{(0)}) \leq \mathbb{E}_q \left[\log \frac{q(x_i^{(1:T)} | x_i^{(0)})}{p_\theta(x_i^{(0:T)} | f, P_r)} \right] = L, \quad (18)$$

according to the process in [67] and considering the set of all shape points containing N points, the above objective can be further adapted to be a series of KL-divergence and entropy terms and the detailed process can be found in Appendix A:

$$L = \mathbb{E}_q \left[\sum_{t=2}^T \sum_{i=1}^N \left(\underbrace{D_{KL}(q(x_i^{(t-1)} | x_i^{(t)}, x_i^{(0)}) || p_\theta(x_i^{(t-1)} | x_i^{(t)}, f, P_r))}_{L_{t-1}} \right) + \underbrace{D_{KL}(q(X^{(T)} | X^{(0)}) || p_\theta(X^{(T)}))}_{L_T} - \sum_{i=1}^N \underbrace{\log p_\theta(x_i^{(0)} | x_i^{(1)}, f, P_r)}_{L_0} \right]. \quad (19)$$

Finally, in order to be consistent with the simplified option proposed in [65], we reparameterize our final refinement objective function by reducing Eq. (19) to an L_2 loss:

$$L_{refine} = \mathbb{E} \left[\sum_{i=1}^N ||\varepsilon - \varepsilon_\theta(x_i^{(t)}, f, P_r, t)|| \right]. \quad (20)$$

Similarly, our pre-training objective function can be simply modified to learn a conditional generative model given prior shapes. We modifies this generative model by conditioning the step function ε_θ on the input tensor that combines information derived from both the current state $x_i^{(t)}$ and the shape prior P_r :

$$L_{pretrain} = \mathbb{E} \left[\sum_{i=1}^N ||\varepsilon - \varepsilon_\theta(x_i^{(t)}, P_r, t)|| \right]. \quad (21)$$

3.5 Self-Supervised Pose and Size Estimation

Having the final trained model while the timestep t is set to zero, we introduce two self-supervised estimation branches:

6-DoF Poses Estimation Branch. The objective of this branch is to develop an estimation decoder from the

Algorithm 1 Self-Supervised Training

Require: T : total timesteps;
Require: P_r : corresponding shape prior sets;
Require: P_0 : original observable point cloud;
Require: $D_{data}(P_0)$: partial point cloud dataset;

① Pre-training Phase:

- 1: **repeat**
- 2: Sample $X^{(0)} \sim D_{data}(P_0) \rightarrow x_i^{(0)}$;
- 3: Sample $t \sim \text{Uniform}(\{1, 2, \dots, T\})$;
- 4: Sample $\varepsilon \sim \mathcal{N}(0, I)$ and β_t ;
- 5: Sample β_t ;
- 6: Compute $\tilde{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$;
- 7: Compute $x_i^{(t)} = \sqrt{\tilde{\alpha}_t} x_i^{(0)} + \sqrt{1 - \tilde{\alpha}_t} \varepsilon$;
- 8: Perform gradient descent with Eq. (21);

$$\nabla_\theta \left(\sum_{i=1}^N ||\varepsilon - \varepsilon_\theta(x_i^{(t)}, P_r, t)|| \right);$$

9: **until** converged

② Refinement Phase:

- 10: **repeat**
- 11: Sample $X^{(0)} \sim D_{data}(P_0) \rightarrow x_i^{(0)}$;
- 12: Sample $t \sim \text{Uniform}(\{1, 2, \dots, T\})$;
- 13: Sample $\varepsilon \sim \mathcal{N}(0, I)$;
- 14: Sample $x_i^{(t)}$ from previous phase;
- 15: Sample P_0 ;
- 16: Learn shape/observation latent f from P_0 and P_r ;
- 17: Perform gradient descent and fine-tune with Eq. (20);

$$\nabla_\theta \left(\sum_{i=1}^N ||\varepsilon - \varepsilon_\theta(x_i^{(t)}, f, P_r, t)|| \right);$$

18: **until** converged

per-group features map $F^{(l)}$ to the SE(3)-equivariant 6-DoF poses for each visible objects. Initially, we aim to predict a subset of pose assumptions including every 6-DoF pose hypothesis for corresponding icosahedron group within $\{G_g\}$. Here, this estimation decoder maps $F^{(l)} \in \mathbb{R}^{(N_0/2^{l+1}) \times d \times |G_g|}$ to $\{\tilde{p}^{(i)}\} \in \mathbb{R}^{K \times (4+3) \times |G_g|}$, ($i = 1, 2, \dots, |G_g|$), and here 4 represents the rotation vector $\tilde{R}_k^{(i)}$ acted by a quaternion $\tilde{q}_k^{(i)}$ and 3 denotes the translation vector $\tilde{t}_k^{(i)}$. To implement this goal, a channel-wise MLP is applied: $f \in \mathbb{R}^{N_0/2^{l+1}} \rightarrow f \in \mathbb{R}^K$, operating solely on each group features map $F^{(l)}$ individually, yielding a new set of features. Then, the feature-wise max-pooling over $\{F^{(l)}\}$ is performed, followed by an MLP-based pose decoder to obtain the final pose assumption subset $\{\tilde{p}^{(i)}\}$.

3D Sizes Recovery Branch. The objective of this branch is to learn an SE(3)-invariant 3D sizes recovery decoder for every observable objects, capable of predicting the per-object's size from the feature map F_4 . Initially, the recovery function maps F_4 to another size assumptions subset $\{\tilde{s}^{(j)}\} \in \mathbb{R}^{K \times 1 \times |G_g|}$, ($j = 1, 2, \dots, |G_g|$) containing each scale at each object for corresponding icosahedron group within $\{G_g\}$, where 1 denotes the value for per-object 3D scale $\tilde{s}_k^{(j)}$. To achieve 3D scale-invariant, both feature map F_4 and the corresponding feature after zero convolution are concatenated, as shown in the top right of Fig. 2. This

Algorithm 2 Inference

Input: total timesteps: T ;
Input: shape prior point: P_r ;
Input: current observable point cloud: P_0 ;
Input: noise distribution: $X^{(T)} \leftarrow \{x_i^{(T)}\} \sim \mathcal{N}(0, I)$;
Output: 6-DoF poses: $\tilde{P}_k = \{\tilde{R}_k, \tilde{t}_k\}$ and 3D scales: $\tilde{s}_k$;
Output: posed shapes after reconstruction: $P = \{P_k\}_{k=1}^K$;

*** Canonical Shapes Reconstruction:**

- 1: **for** $t = T, T-1, \dots, 1$ **do**
- 2: Sample $z \sim \mathcal{N}(0, I)$ if $t > 1$ else $z = 0$;
- 3: Sample β_t and $\alpha_t = 1 - \beta_t$;
- 4: Compute $\tilde{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$ and $\sigma_t = \frac{1 - \tilde{\alpha}_t}{1 - \tilde{\alpha}_1} \beta_t$;
- 5: Compute points distribution in each timestep:

$$(x_i^{(t-1)}, P_r) = \frac{1}{\sqrt{\alpha_t}}(x_i^{(t)} - \frac{\beta_t}{\sqrt{1 - \tilde{\alpha}_t}} \varepsilon_\theta(x_i^{(t)}, f, P_r, t)) + \sigma_t z;$$

- 6: **end for**
- 7: Sample $\tilde{X}^{(0)} = \{x_i^{(0)}, P_r\}$ and $\tilde{X}^{(0)} = \{\tilde{X}_k^{(0)}\}_{k=1}^K$;

*** Pose/size Estimation and Shapes Recovery:**

- 8: Set timestep $t = 0$;
- 9: Recover the assumption subsets $\{\tilde{P}^{(i)}\}$ and $\{\tilde{s}^{(j)}\}$;
- 10: **for** $k = 1, 2, \dots, K$ **do**
- 11: **for** $i = 1, 2, \dots, |G_g|$ **do**
- 12: **for** $j = 1, 2, \dots, |G_g|$ **do**
- 13: Pose $P_k^{(i,j)}$ with Eq. (22);
- 14: **end for**
- 15: **end for**
- 16: **end for**
- 17: Calculate the distance $d(P_k^{(i,j)}, P_0^k)$ with Eq. (23);
- 18: Select optimum $\tilde{P}_k^{(i^*)}$, $\tilde{s}_k^{(j^*)}$ and $P_k^{(i^*, j^*)}$ with Eq. (24);
- return** $\tilde{P}_k$, $\tilde{s}_k$ and P .

concatenated output undergoes similar operations in the next three stages, along with an MLP, to obtain a base feature map with the size of $K \times d$. Subsequently, this process is repeated and added to the generated features from the pose estimation branch, and an MLP-based size decoder is used to obtain the final size assumption subset $\tilde{s}^{(j)}$.

Given both pose and size assumption subsets, we can pose the canonical shapes point cloud using each assumption $\tilde{P}_k^{(i)}$ and $\tilde{s}_k^{(j)}$, rendering posed reconstruction shape points for per-object and every hypotheses:

$$P_k^{(i,j)} = \tilde{s}_k^{(j)} \cdot \tilde{R}_k^{(i)} \cdot \tilde{X}_k^{(0)} \cdot (\tilde{R}_k^{(i)})^{-1} + \tilde{t}_k^{(i)}. \quad (22)$$

This process involves calculating the point cloud distance between them and the original input of partial observation $P_0 = \{P_0^k\}_{k=1}^K$:

$$d(P_k^{(i,j)}, P_0^k) = \sum_{k=1}^K \left(\sum_{x \in P_k^{(i,j)}} \min_{y \in P_0^k} \|x - y\|^2 + \sum_{y \in P_0^k} \min_{x \in P_k^{(i,j)}} \|x - y\|^2 \right), \quad (23)$$

we select the best pose and size assumption values, and corresponding posed shape points as the final results, that

satisfy the minimum distance:

$$(i^*, j^*) = \arg \min_{i, j \in \{1, 2, \dots, |G_g|\}} d(P_k^{(i,j)}, P_0^k). \quad (24)$$

Further details on inference process for pose and size can be found in the last part of Alg. 2.

4 EXPERIMENTS

In this section, we conduct extensive experiments to evaluate the performance of our method and compare it with currently available state-of-the-art baselines on five datasets, including four public datasets and our self-built dataset.

4.1 Experimental Setup

Datasets. The REAL275 and CAMERA25 datasets [8] consist of six different categories: *bottle*, *bowl*, *camera*, *can*, *laptop* and *mug*. The CAMERA25 dataset contains a total of 300K synthetic RGB-D images generated by rendering and compositing virtual instances from ShapeNetCore [74]. Approximately 90% of the images, covering 1,085 instances, are divided into the training dataset while the remaining 10% (25K) covering 184 different instances are reserved for evaluation. The REAL275 dataset contains real-world RGB-D images, with 4,300 images for training and 2,750 images for evaluation. The training set is collected from seven scenes, and the testing set contains 6 scenes with 3 unseen instances per category. The YCB-Video dataset [20] is a challenging dataset characterized by strong occlusion, clutter and several symmetric objects. It contains 21 objects and 92 video sequences. The Wild6D dataset [19] is a large-scale RGB-D dataset, including 5,166 videos covering 1722 different object instances across 5 categories. We employ the same train and test sets as [19]. Omni6DPose is a substantial dataset with large scale object categories under occlusion and clutter, and we only use its real component, ROPE.

Our self-built dataset, termed "DYNAMIC45", was collected using an air-ground robotic system to capture RGB-D data in dynamic scenes (see left of Fig. 9). These data were captured by a RGB-D camera sensor (RealSense D435i) and contains 16 videos with 8 scenes and 10 categories, totaling 45K images. We split these videos into 12 for training and 4 for testing. To automatically annotate these frames, we first calibrate the camera extrinsic matrix for each video and then use an offline 3D labeling tool to obtain the necessary annotations. This dataset will be made public.

Metrics. Following [45], we evaluate the performance of 6-DoF pose/size estimation and 3D shape recovery using four types of metrics: (1) *3D Intersection over Union* (3D-IoU), measures the average accuracy for various IoU-overlap thresholds (we take 50% and 75% as the thresholds). (2) *a°b cm*, measures the fraction of the rotation error less than a and the translation error less than b for each instance (we take $5^\circ 2$ cm, $5^\circ 5$ cm, $10^\circ 2$ cm and $10^\circ 5$ cm as the thresholds). (3) On YCB-Video dataset, we employ extra commonly used metrics, including ADD (S) and the AUC (area under curve) of ADD-S and ADD(-S), similar to [2]. (4) For 3D shape reconstruction, we adopt the *Chamfer Distance* (CD), that calculates the average closest point distance, following [10].

TABLE 2

Quantitative comparison of fully-supervised category-level pose estimation on the public CAMERA25 and REAL275 dataset. Note that the best and the second best results are highlighted in bold and underlined. The comparison results of current state-of-the-art baselines are summarized from their original papers. Empty entries indicate that they were not reported in the original paper.

Method	CAMERA25 Dataset						REAL275 Dataset					
	<i>IoU</i> 50	<i>IoU</i> 75	5°2cm	5°5cm	10°2cm	10°5cm	<i>IoU</i> 50	<i>IoU</i> 75	5°2cm	5°5cm	10°2cm	10°5cm
1 NOCS [8] [CVPR2019]	83.9	69.5	32.3	40.9	48.2	64.6	80.5	30.1	7.2	10.0	13.8	25.2
2 SDP [10] [ECCV2020]	93.2	83.1	54.3	59.0	73.3	81.5	77.3	53.2	19.3	21.4	43.2	54.1
3 FS-Net [38] [CVPR2021]	-	85.2	-	62.0	-	60.8	81.1	52.0	19.9	33.9	-	69.1
4 SGPA [9] [ICCV2021]	93.2	88.1	70.7	74.5	82.7	88.4	80.1	61.9	35.9	39.6	61.3	70.7
5 CenterSnap [45] [ICRA2022]	92.5	-	-	66.2	-	81.3	80.2	-	-	27.2	-	58.8
6 ShAPO [68] [ECCV2022]	93.5	-	-	66.6	-	81.9	79.0	-	-	48.8	-	66.8
7 SAR-Net [42] [CVPR2022]	86.8	79.0	66.7	70.9	75.3	80.3	79.3	62.4	31.6	42.3	50.3	68.3
8 GPV-Pose [41] [CVPR2022]	92.9	86.6	67.4	76.2	-	87.4	<u>83.0</u>	64.4	32.0	42.9	-	73.3
9 HS-Pose [69] [CVPR2023]	93.3	89.4	73.3	80.5	80.4	89.4	82.1	74.7	46.5	55.2	68.6	82.7
10 IST-Net [70] [ICCV2023]	<u>93.7</u>	<u>90.8</u>	71.3	79.9	79.4	89.9	82.5	<u>76.6</u>	47.5	53.4	<u>72.1</u>	80.5
11 Query6DoF [11] [ICCV2023]	91.9	88.1	78.0	83.1	83.9	90.0	82.5	76.1	46.8	54.7	67.9	81.6
12 VI-Net [71] [ICCV2023]	-	79.1	74.1	81.4	<u>79.3</u>	<u>87.3</u>	-	48.3	<u>50.0</u>	<u>57.6</u>	70.8	<u>82.1</u>
13 GPT-COPE [72] [TCSVT2023]	92.5	86.9	70.4	76.5	81.3	88.7	82.0	70.4	45.9	53.8	63.1	77.7
14 MH6D [73] [TNNLS2024]	-	-	-	-	-	-	70.2	45.0	40.4	50.1	58.3	73.1
Ours	94.5	92.1	79.4	85.2	88.5	92.0	86.1	77.1	55.0	59.8	73.7	83.7

TABLE 3

Quantitative comparison of self-supervised category-level pose estimation on the public REAL275 dataset. Note that the best and the second best results are highlighted in bold and underlined. The comparison results of other baselines are summarized from their original papers, and empty entries indicate that they were not reported. "Syn." and "Real" denote the uses of training data of synthetic CAMERA25 and real-world REAL275 datasets, respectively. "Multiple" and "Single" indicate whether the method is for the single object or for multiple objects.

Method	Training Data			Objective		Shape Prior	Evaluation Metrics					
	Input	Real	Syn.	Multiple	Single		<i>IoU</i> 50	<i>IoU</i> 75	5°2cm	5°5cm	10°2cm	10°5cm
SSC-6D [14] [AAAI2022]	RGB-D	✓	✓		✓	✓	72.9	-	16.8	19.6	44.12	54.5
Self-DPDN [13] [ECCV2022]	RGB-D	✓	✓		✓	✓	<u>83.0</u>	<u>70.3</u>	<u>39.4</u>	<u>45.0</u>	<u>63.2</u>	<u>72.1</u>
UDA-COPE [12] [CVPR2022]	RGB-D	✓	✓		✓	✗	82.6	62.5	30.4	34.8	56.9	66.0
Zaccaria et al. [17] [RAL2023]	RGB-D	✓	✓		✓	✗	75.2	-	34.9	39.5	53.1	63.8
Self-Pose [18] [ICLR2023]	RGB	✓	✗		✓	✓	41.7	-	-	11.6	-	28.3
SCNet [26] [T-MECH2023]	RGB-D	✓	✓		✓	✓	74.7	30.9	21.7	23.5	47.6	64.9
Ours	D	✓	✓	✓		✓	86.1	77.1	55.0	59.8	73.7	83.7

Implementation Details. Our network is implemented using PyTorch and trained in REAL275 and CAMERA25. We train our model using the Adam Optimizer with initial learning rate of $1e^{-4}$ (decayed by 0.7 every 40 epochs) and a batch size of 32 on an NVIDIA RTX A6000 GPU. To ensure fairness, we also utilize the available segmentation results generated by an off-the-shelf network (Mask-RCNN) to crop all objects from the RGB-D frame by using camera intrinsic parameters and recover the partial observable point cloud P_0 . We pre-sample the points in $D_{data}(P_0)$ such that $N_0 = N$. The number of points in each shape prior is set as 2,048. The number of timesteps T in the diffusion process is 100, and we set the variance schedules from $\beta_1 = 10^{-4}$ to $\beta_T = 0.5 \times 10^{-1}$, with β_t being linearly interpolated.

4.2 Comparison and Evaluation on Public Datasets

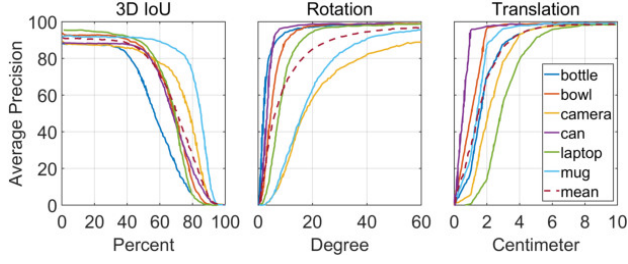
4.2.1 Fully-Supervised Category-Level Pose Estimation

We first compare the performance of our approach with state-of-the-art category-level pose estimation methods trained under the full supervision of pose annotations. As shown in Table 2, we compare our results with 14 available full-supervised methods on both CAMERA25 and REAL275 datasets. Our final refined model achieves state-of-the-art results for all evaluation metrics on both two dataset. Notably, on the CAMERA25 dataset, we outperforms Query6DoF [11], the current SoTA method, by

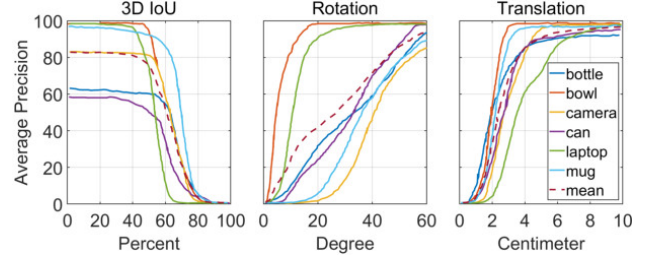
4.0% and 4.6% in terms of *IoU*75 and 10°2cm. Similarly, on another real-world REAL275 dataset, we also outperform other available baselines. Specifically, our method exhibits superior performance, achieving an mAP of 86.1% for *IoU*50, 55.0% for 5°2cm and 59.8% for 5°5cm. These results indicate that the robustness performance of our diffusion-based self-supervised learning. Furthermore, we report comparative results of the category-specific mAP (mean average precision) between our method and NOCS [8] across both public datasets. Detailed results can be found in Fig. 5, where our method outperforms the NOCS by a large margin across most categories in both datasets. These experiments show that our method's superior performance for category-level pose estimation.

4.2.2 Self-Supervised Category-Level Pose Estimation

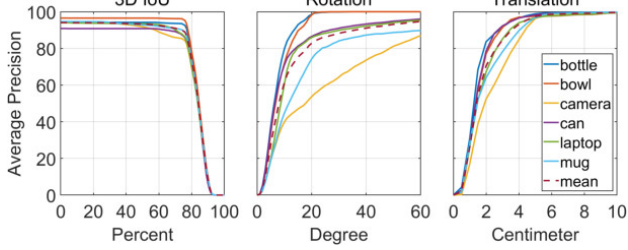
We conducted some experiments to compare our method with existing published category-level self-supervised methods on REAL275 dataset, including those without the use of shape prior [12], [17] and those utilizing shape prior [13], [14], [18], [26]. It's worth noting that our method is currently the only one for multi-object pose estimation among these self-supervised approaches. As depicted in TABLE 3, we outperform others by a significant margin across all evaluation metrics. In detail, we achieve 55.0% and 59.8% on 5°2cm and 5°5cm, respectively, surpassing Self-DPDN [13], the current most powerful method, by 15.6%



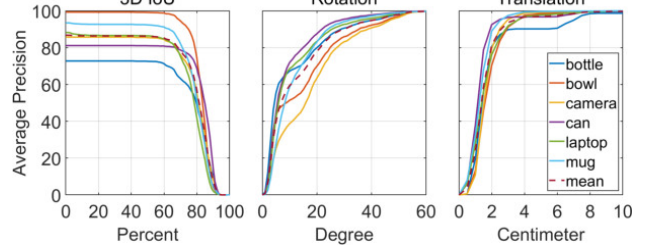
(a) Baseline (NOCS) results on CAMERA25 dataset.



(b) Baseline (NOCS) results on REAL275 dataset.



(c) Our results on CAMERA25 dataset.



(d) Our results on REAL275 dataset.

Fig. 5. Comparison of mAP on both CAMERA25 and REAL275 datasets. Mean average precision (mAP) of our method and baseline for various 3D IoU, rotation and translation thresholds on CAMERA25 and REAL275 datasets. The upper row (*i.e.*, (a) and (b)) is the results of NOCS and all results are from original paper [8], and the bottom row (*i.e.*, (c) and (d)) is ours.

TABLE 4

Quantitative comparison of category-level pose estimation on the public Wild6D dataset. Note that the results of SSC-6D [14] and Self-DPDN [13] are from our test using their public codes, and other results are from Self-Pose [18]. Column of Data indicates the training data: "C=CAMERA25", "R=REAL275" and "W=Wild6D". "Full." and "Self." represent the learning scheme.

Method	Scheme	Data	Evaluation Metrics				
			IoU_{50°	$5^\circ 2cm$	$5^\circ 5cm$	$10^\circ 2cm$	$10^\circ 5cm$
CASS [43]	Full.	C+R	1.0	0	0	0	0
SPD [10]		C+R	32.5	2.6	3.5	9.7	13.9
DualNet [39]		C+R	70.0	17.8	22.8	26.3	36.5
GPV-Pose [41]		C+R	67.8	14.1	21.5	23.8	41.1
RePoNet [19]		C+W	<u>70.3</u>	29.5	34.4	35.0	42.5
SSC-6D [14]	Self.	C+R	68.2	25.2	30.9	33.9	39.6
Self-DPDN [13]		C+R	69.5	31.6	36.3	40.0	43.2
Self-Pose [18]		W	68.2	<u>32.7</u>	35.3	38.3	<u>45.3</u>
Ours	Self.	C+R	71.1	33.3	38.5	43.6	48.4
Ours (+ Wild6D)		C+R+W	73.0	35.2	39.7	46.4	50.9

and 14.8%. Regarding prior-free methods, compared with UDA-COPE [12], we demonstrate substantial improvements across all metrics, *e.g.*, 77.1% vs. 62.5% on IoU_{75} , 55.0% vs. 30.4% on $5^\circ 2cm$ and 59.8% vs. 34.8% on $5^\circ 5cm$. Qualitative visualization comparisons between several state-of-the-art (SoTA) approaches and our method across different scenes in REAL275 and CAMERA25 dataset are shown in Fig. 6.

Moreover, to assess our method's generalization performance, we present the results on the public Wild6D dataset. We compare with both fully-supervised and self-supervised category-level methods by using our trained model on REAL275 and CAMERA25 and our model with fine-tuning on Wild6D dataset, respectively. The results are reported in Table 4. It is evident that our unfine-tuned model outperforms other available baselines, and the corresponding fine-tuned model exhibits an obvious improvement across various metrics. Therefore, we claim that our proposed algorithm is more effective due to its

TABLE 5

Quantitative comparison of instance-level pose estimation on the public YCB-Video dataset. The comparison results of other baselines are summarized from their original papers, and empty entries indicate that they were not reported. Column of Supervision indicates the training manner: "Syn.=synthetic data", "Real=real-world data" and "GT=ground-truth". "Semi." and "Self." represent the learning scheme.

Method	Supervision	Evaluation Metrics		
		ADD (S)	AUC of ADD-S	AUC of ADD (-S)
PoseCNN [20]	Syn. + Real GT	21.3	75.9	61.3
PVNet [31]		-	-	73.4
DeepIM [75]		-	88.1	81.9
CosyPose [76]		-	89.8	<u>84.5</u>
SO-Pose [23]		56.8	90.9	83.9
Zhou et al. [6]	Real+ Semi.	-	90.4	-
Self6D++ [2]	Syn. + Self.	-	<u>91.1</u>	80.0
Hai et al. [4]	Real+ Self.	<u>67.4</u>	-	-
Ours	Syn.+ Real+ Self.	70.7	92.0	85.3

utilization of a diffusion-based self-learning scheme with the SE(3)-equivariant and scale-invariant representation. Fig. 7 (b) presents some qualitative results on Wild6D.

4.2.3 Self-Supervised Instance-Level Pose Estimation

Table 5 presents a comparison of our method with various instance-level approaches on YCB-Video dataset, using the common metric of ADD(S) and the extra standard metric AUC of ADD-S/ADD(-S). We compare not only with available instance-level approaches under the fully-supervision setting (row-1 to row-5 in Table 5), but also with state-of-the-art baselines under the self-supervised [2], [4] or semi-supervised [6] manner. Although our method is not specifically designed for these instance-level self-supervised pose estimation, such as Self6D++ [2], our model still achieves robust performance, compared to or even surpassing the state-of-the-arts. Notably, our approach shows improved results for the ADD(S) metric with 70.7%

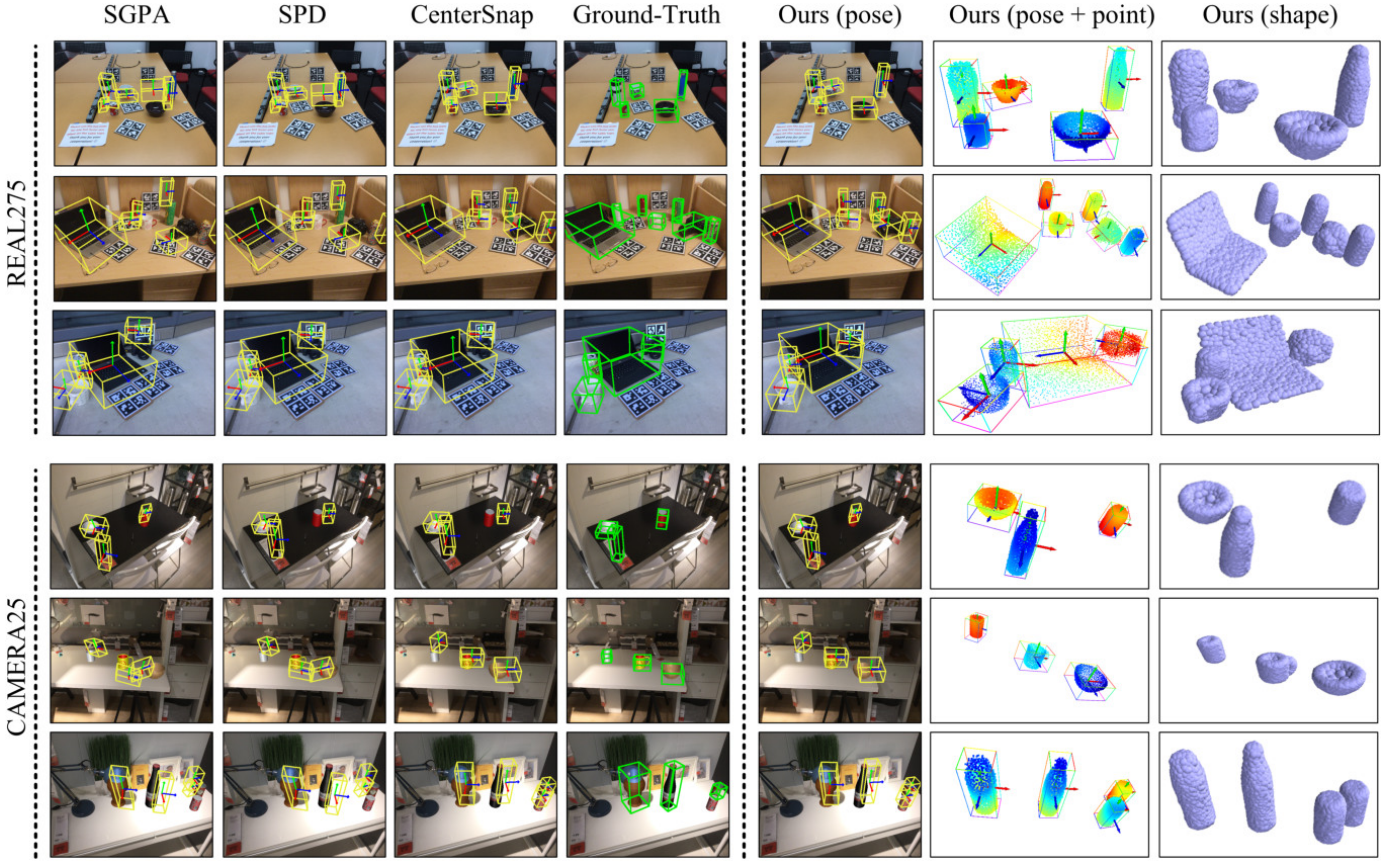


Fig. 6. Visualization comparison results on both REAL275 and CAMERA25 datasets. We compare our method with other representative SoTA approaches in terms of 6-DoF pose and size estimation. Meanwhile, we also show the corresponding result of 3D shape reconstruction (right side). The yellow and green boxes represent the predictions and ground-truth labels, respectively.

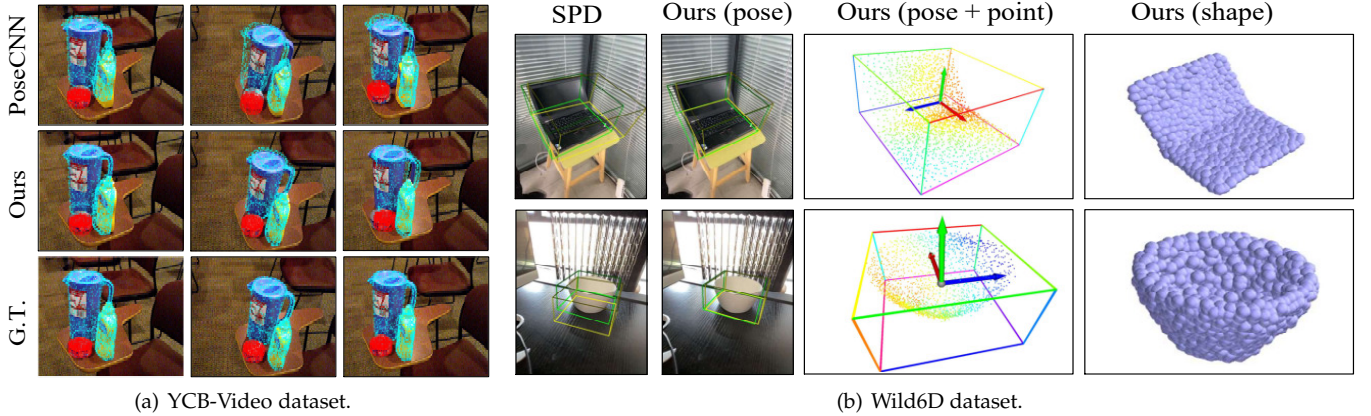


Fig. 7. Visualization comparison results on both YCB-Video and Wild6D datasets. (a) Qualitative results of our method and PoseCNN [20] on YCB-Video dataset. To keep in line with PoseCNN, each object shape model are transformed with the predicted pose and then projected into the 2D images. (b) Qualitative results of our method and SPD [10] on Wild6D dataset. Yellow and green represent predicted and ground-truth results.

compared to 67.4% reported in [4], and achieves slightly better performance for the AUC of ADD-S and ADD(-S) with 92.0% vs. 91.1% and 85.3% vs. 80.0% from Self6D++ [2]. We also provide qualitative results on YCB-Video dataset, displayed in Fig. 7 (a).

4.2.4 3D Shape Reconstruction Evaluation

To evaluate the quality of shape reconstruction, we first compare our method with 10 state-of-the-art baselines using the metric of *CD*, that calculates the distance between

our reconstructed shape model P and the ground truth shape model $\hat{P}$ in the canonical space. As reported in Table 6, our method obtains average *CD* metrics of 0.10 on the CAMERA25 dataset and 0.06 on the REAL275 dataset, outperforming the current well-performed baseline, ShAPO [68] by 0.06 and 0.07, respectively. Notably, among these baselines, SSC-6D [14] is the only self-supervised approach, and our model outperforms it across the majority of categories. The main reason for this case is that these baselines tend to produce a mean shape for

TABLE 6

Quantitative comparison of 3D shape reconstruction on public CAMERA25 and REAL275 dataset: Evaluated with CD metric (10^{-2}). Note that the best and the second best results are highlighted in bold and underlined, and the lower the better. The comparison results of current state-of-the-art baselines are summarized from their original papers. Empty entries were not reported in the original paper.

Method	CAMERA25 Dataset							REAL275 Dataset						
	Bottle	Bowl	Camera	Can	Laptop	Mug	Mean	Bottle	Bowl	Camera	Can	Laptop	Mug	Mean
1 CASS [43] [CVPR2020]	-	-	-	-	-	-	-	0.17	0.09	0.53	0.18	0.19	0.24	0.23
2 Reconstruction [10] [ECCV2020]	0.18	0.16	0.40	0.09	0.20	0.14	0.20	0.34	0.12	0.89	0.15	0.29	<u>0.10</u>	0.32
3 SPD [10] [ECCV2020]	0.17	0.15	0.42	0.09	0.20	0.14	0.20	0.34	0.12	0.89	0.15	0.29	<u>0.10</u>	0.32
4 C3R-Net [37] [IROS2021]	0.13	0.10	0.26	<u>0.08</u>	<u>0.09</u>	<u>0.11</u>	<u>0.13</u>	0.30	0.10	0.76	0.13	0.13	<u>0.12</u>	0.26
5 SGPA [9] [ICCV2021]	0.13	0.13	0.33	<u>0.08</u>	0.12	<u>0.11</u>	0.15	0.29	0.09	0.55	0.17	0.16	0.11	0.23
6 CenterSnap [45] [ICRA2022]	<u>0.11</u>	0.10	0.29	0.13	0.07	<u>0.12</u>	0.14	0.13	0.10	0.43	0.09	<u>0.07</u>	0.06	0.15
7 SSC-6D [14] [AAAI2022]	-	-	-	-	-	-	-	0.06	0.04	0.38	0.10	0.03	0.45	0.18
8 6D-ViT [44] [TIP2022]	0.14	<u>0.12</u>	0.29	0.09	0.12	0.12	0.15	0.24	0.11	0.61	0.16	0.14	0.11	0.21
9 ShAPO [68] [ECCV2022]	0.14	0.13	<u>0.20</u>	0.14	0.07	<u>0.11</u>	0.16	0.10	0.08	0.40	<u>0.07</u>	0.08	0.06	<u>0.13</u>
10 GCASP [77] [PMLR2023]	-	-	-	-	-	-	-	0.21	0.16	<u>0.11</u>	<u>0.16</u>	0.21	0.29	0.19
Ours	0.10	0.10	0.17	0.07	0.07	0.08	0.10	0.06	<u>0.07</u>	0.09	0.06	0.03	0.06	0.06

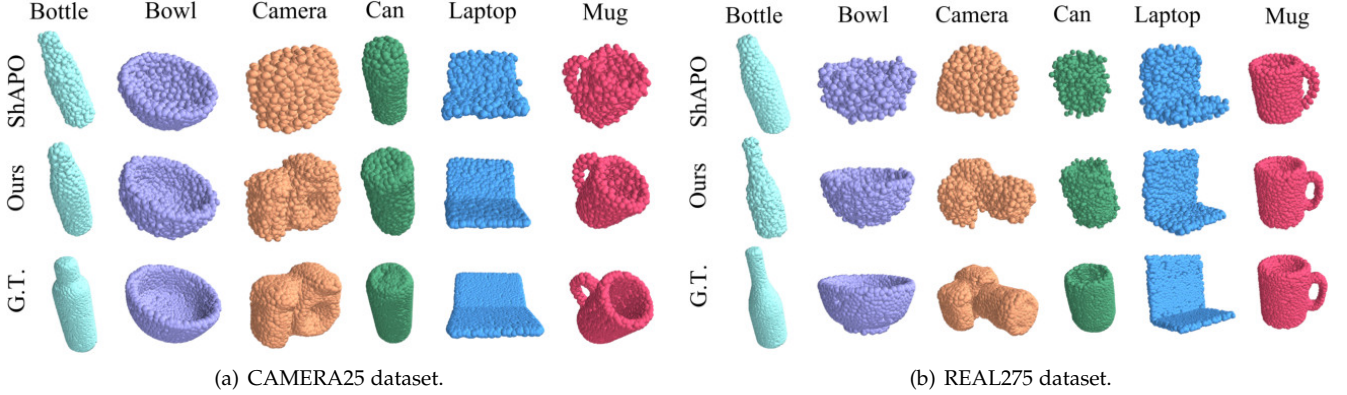


Fig. 8. Visualization comparison results of 3D shapes reconstruction. We compared our proposed method with the strong currently available baseline (ShAPO [68]) for all six categories on both CAMERA25 and REAL275 datasets. It is clear that our results are closer to the ground-truth. To facilitate comparison, we align all results with same category into a similar pose.

TABLE 7

Quantitative comparison of 3D shape reconstruction on public YCB-Video dataset: Evaluated with CD metric (10^{-2}). Note that all reported results are from models using their public codes or checkpoints and fine-tuned on YCB-Video for a fair comparison.

Object	Fully-Supervised			Self-Supervised	
	SGPA [9]	SPD [10]	ShAPO [68]	SSC-6D [14]	Ours
004 sugar box	0.15	0.20	<u>0.11</u>	<u>0.11</u>	0.10
019 pitcher base	0.24	0.31	<u>0.17</u>	0.18	0.14
021 bleach cleanser	0.30	0.33	0.27	<u>0.20</u>	0.17
036 wood block	<u>0.09</u>	0.11	<u>0.09</u>	0.07	0.07
040 large marker	<u>0.14</u>	0.18	0.22	0.16	0.10
Mean	0.18	0.23	0.17	<u>0.14</u>	0.12

each instance within the same category, however, our method tries to seek the complete shape from partially observable point clouds, that may be closer to the ground-truth than the mean shape. This distinction is evident in the visualization comparison results displayed in Fig. 8. Furthermore, to verify our proposed method's shape reconstruction performance for different shape categories, we conduct an additional experiment on the YCB-Video dataset, considering five new unseen object categories such as box, etc. As depicted in Table 7, we compared our method with several fully-supervised approaches (SGPA [9], SPD [10] and ShAPO [68]), as well as the

TABLE 8

Robustness comparison of large-scale Categories on public Omni6dPose (ROSE) dataset: Evaluated with CD metric (10^{-2}). Note that all results are from models using their public checkpoints.

Method	Pose and Size					Shape CD
	IoU_{50}	$5^{\circ}2cm$	$5^{\circ}5cm$	$10^{\circ}2cm$	$10^{\circ}5cm$	
SPD [10]	2.5	5.5	7.3	10.2	13.2	0.56
SGPA [9]	2.0	4.3	6.7	9.3	15.0	0.63
CenterSnap [45]	12.4	10.7	15.4	17.0	21.3	0.44
Ours	22.3	12.2	18.9	24.7	30.6	0.30

currently available self-supervised baseline SSC-6D [14], respectively. On this benchmark dataset, our method also achieves the state-of-the-art results. From these comparative results and the visualization of category-specific shape reconstruction, it is evident that our self-supervised network effectively enhances the quality of 3D shape reconstruction. This further indicates that our proposed diffusion-driven pretrain-to-refine self-supervised learning strategy not only can improve the performance of pose and 3D size estimation, but also facilitates corresponding 3D shape reconstruction.

4.2.5 Robustness Comparison of Large-scale Categories

To validate our method for large-scale object categories under the occlusion and clutter conditions, we also compare

TABLE 9

Comparison of computational efficiency and inference time. Note that all results are from models using their public checkpoints.

Metric	SPD [10]	SGPA [9]	Ours w/o RK-Point	Ours w/o 3DSI-Graph	Ours w/o SE(3)-Group	Ours (full)
Params (M)	18.30	23.31	14.55	14.06	15.78	20.44
Inference (ms)	65.66	70.84	52.39	52.42	53.87	55.67

pose estimation and shape reconstruction performance in public dataset Omni6dPose (ROSE), that is a substantial real-world dataset with large-scale object categories, and the results are presented in Table. 8. We can see that our self-supervised network obtains the best scores in all evaluation metrics, which is a direct certification of our robustness.

The underlying reason for these observed gains is that the proposed SE(3) equivariance facilitates learning rotation-equivariant correlation aligned with object-body transformations, enabling more robust generalization across unseen viewpoints. Besides, the diffusion process serves not merely as shape reconstruction mechanism but as our self-supervised two stage training pipeline, which can effectively combine with equivariant features to enhance generalization to pose and shape.

4.2.6 Computational Efficiency Comparison

Table. 9 provides the detailed comparison for computational efficiency and inference time among existing methods (SPD [10], SGPA [9]) and various ablation versions of our proposed approach on NOCS-REAL275 dataset. It is noted that all results are tested using the same system: NVIDIA RTX A6000 GPU. Despite slight more complexity, our full model achieves a reasonable inference time and ablation variants are even faster, showing that our full model strikes a favorable balance between accuracy and efficiency.

4.3 Ablation Studies and Analysis

4.3.1 Evaluation of Different Components

To validate the effectiveness of the specific component design in our ORT module, we conduct the detailed ablation experiments on REAL275 dataset, as depicted in Table. 10. The last row in the table represents the complete ORT module, while the first three rows display the results of ablating the point convolution layer with radial-kernels (RK-Point layer), the 3D scale-invariant graph convolution layer (3DSI-Graph layer) and the SE(3) group convolution layer (SE(3)-Group layer), respectively. Additionally, we examined the effect of our seed points aggregation and the results using the base Transformer model without adding any proposed components (w/o SE(3) block).

1) *Effect of RK-Point layer*: Initially, we investigate the impact of the proposed RK-Point layer on the model performance by removing this layer while remaining other layers. As shown in the row-1 in Table. 10, no RK-Point Layer results in a significant decline in most pose estimation metrics upon ablation, *i.e.*, $IoU50$: 25.6%↓, $5^\circ 2cm$: 15.1%↓ and $10^\circ 5cm$: 18.4%↓, while the shape reconstruction metric remains essentially unchanged. This proves the effectiveness of our radial kernel mapping in selecting SE(3) equivariant pose-aware features.

TABLE 10

Ablation study of our proposed module to investigate each components contribution on REAL275 dataset. Note that 3D shape reconstruction is evaluated with CD metric (10^{-2}).

Method	Pose and Size						Shape CD
	$IoU50$	$5^\circ 2cm$	$5^\circ 5cm$	$10^\circ 2cm$	$10^\circ 5cm$		
Ours (w/o RK-Point)	60.5	39.9	49.2	62.7	65.3		0.07
Ours (w/o 3DSI-Graph)	84.6	52.9	55.3	71.2	81.6		0.24
Ours (w/o SE(3) block)	59.2	38.5	46.1	50.8	62.4		0.30
Ours (w/o seed points)	84.6	52.9	57.5	73.0	82.2		0.08
Ours (w/o SE(3)-Group)	72.4	46.5	50.1	62.8	70.0		0.10
Ours (w/ SE(3) Point Conv)	82.0	54.1	55.6	68.0	79.3		0.12
Ours (w/ Classical ICP)	84.9	52.3	54.6	72.0	77.4		0.06
Ours	86.1	55.0	59.8	73.7	83.7		0.06

2) *Effect of 3DSI-Graph layer*: Next, we verify the effectiveness of our 3DSI-Graph Layer using the same operation as above. The results from row-2 in Table. 10 demonstrate a decrease of 0.18 in the CD metric upon ablating the 3DSI-Graph Layer, with minimal decay observed in pose metrics. This confirms the effectiveness of our proposed scale-invariant graph convolution layer in enhancing the network's ability to capture scale/shape invariance information for shape reconstruction.

3) *Effect of SE(3) block*: The result in row-3 reveals the performance of our simplest model without adding any proposed components, *w.r.t.* the ORT module is a modification of multi-head attention. It's clear to see that incorporating our complete SE(3) block into the ORT module drastically improves the model performance across almost all metrics, compared to row-8. These results further show the important role of our proposed three layers in the effectiveness of ORT module.

4) *Impact of seed points generation*: We also evaluate the impact caused by the seed points generation strategy used in ORT. It can be seen that the performance of our model slightly decline when the seed points are absent. The potential reason is that the neighbor domain space obtained from seed points can effectively establish a link between prior and observation, resulting in more relevant features.

5) *Effect of SE(3)-Group layer*: The third experiment aims to verify the effectiveness of the SE(3) group convolution layer. By comparing the results in row-5 and row-7 of Table. 10, we observed that incorporating this layer into ORT module further improves both pose estimation and shape reconstruction, *e.g.*, $10^\circ 5cm$: 13.7%↑ and CD : 0.04↓. This highlights the effectiveness of the SE(3)-Group Layer. To better explore its benefits for both shape and pose learning, we further replace this layer with the SE(3) point convolution layer presented in [62], which also aims at aggregating point spatial correlations locally under a rotation group element. As shown row-6, our method gets better results with the addition of the SE(3)-group layer.

6) *Effect of pose and size estimation*: Besides, we also compare our self-supervised pose and size estimation module with the classical ICP approach and the results are shown in row-6 in Table. 10. It can be seen that our method leads to a more accurate 3D size and 6-DoF pose, further confirming the advantage of our pose and size estimation modules over classical ICP.

TABLE 11

Comparison with different point sampling manners on REAL275 dataset. Note that 'RS', 'US' and 'FPS' mean traditional random sampling, uniform sampling and farthest point sampling, respectively. 'SFD' and 'SFS' stand for the spatial feature distance and the semantic feature similarity in our SGS.

Sampling	Pose and Size					Shape CD
	<i>IoU</i> 50	5°2cm	5°5cm	10°2cm	10°5cm	
RS	69.7	30.3	42.1	53.6	62.2	0.28
US	70.8	38.5	45.4	55.0	68.7	0.24
FPS	72.2	40.6	53.7	62.5	68.9	0.18
SGS w/o SFD	85.0	48.9	57.5	70.4	81.6	0.09
SGS w/o SFS	83.9	50.0	56.4	68.2	79.2	0.11
SGS (Ours)	86.1	55.0	59.8	73.7	83.7	0.06

TABLE 12

Ablation study of proposed self-supervised training manner on REAL275 dataset. Note that "Full" indicates our complete scheme.

Manner	#	Settings	Pose and Size			Shape CD
			<i>IoU</i> 50	5°2cm	10°5cm	
Training Scheme	V1	w/o zero conv	75.3	50.2	76.6	0.08
	V2	w/o trainable copy	66.0	42.3	58.4	0.10
	V3	directly train base model	65.6	30.1	52.0	0.13
Extra Condition	E1	no task-specific condition	72.3	45.2	66.0	0.14
	E2	only observable latent	80.0	49.8	78.4	0.09
	Full	Ours	86.1	55.0	83.7	0.06

4.3.2 Comparison of Different Point Sampling

In the Prior-Aware Pyramid 3D Point Transformer, we stack four shape-guided sampling modules to obtain multi-scale features and enhance the related feature representation. Here, we study the impact of different sampling manners on model performance. Specifically, we explore five schemes on the REAL275 dataset, including three traditional sampling methods and two variants of our SGS, as displayed in Table. 11. In our SGS variants (*i.e.*, row-4 and row-5), all sampling points come from a single pathway. The results reveal that our SGS achieves the best performance. Relying solely on spatial feature distance or semantic feature similarity may lead to difficulties in preserving relevant points and could even result in sampling points clustering around the same instance. In this regard, we choose our proposed complete SGS approach.

4.3.3 Effectiveness of Self-Supervised Training Manner

To investigate the effectiveness of the proposed pretrain-to-refine self-supervised training strategy, we conduct five ablation experiments, considering two situations: training schemes and extra task-specific condition settings.

- V1 "*w/o zero conv*": The zero convolution is replaced by standard convolution with Gaussian weights.
- V2 "*w/o trainable copy*": Each trainable copy of ORT is replaced by single convolution at refinement stage.
- V3 "*directly train base model*": The network is only pre-trained, without the refinement learning phase.
- E1 "*no task-specific condition*": During the refinement phase, no additional conditions are added.
- E2 "*only observable latent*": In the second phase, only observable latent embeddings are added.

The comparison results on REAL275 are summarized in Table. 12, by comparing the performance of these variants, it is obvious that each design plays a positive role. Particularly, comparing "V1", "V2" and "V3", the refinement training process improves all performances. The reason is that the trainable copy of pretrained backbone enables learning the correlation between priori shapes and observations. Besides, compared to the variants "E1" and "E2", the shape/observable latent embedding f improves estimation accuracy. Ultimately, our complete scheme "Full" proves the proposed training strategy is effective for achieving the self-supervised pose estimation and shape reconstruction.

TABLE 13

Statistical comparisons of 6-DoF pose estimation and 3D shape reconstruction on our self-built dataset. We compare it with state-of-the-art baseline (CenterSnap [45]). The "std" is standard deviation, and the results of CenterSnap are based on its official checkpoint and code from original paper.

Method	Category	<i>IoU</i> 50 ↑		5°5 cm ↑		10°5 cm ↑		CD ↓	
		mean	std	mean	std	mean	std	mean	std
CenterSnap [45]	Bowl	86.2	3.03	11.4	2.78	28.0	0.21	0.12	0.91
	Bottle	89.8	4.17	9.8	3.99	25.7	0.39	0.23	0.56
	Can	89.7	2.42	15.4	3.31	33.6	0.57	0.15	0.34
	Mug	90.0	1.47	14.8	0.57	18.6	0.44	0.15	0.28
	Laptop	86.9	1.47	16.9	0.57	25.3	0.48	0.27	0.24
Ours	Bowl	99.3	2.31	25.7	1.63	59.9	0.33	0.09	0.65
	Bottle	98.3	2.16	29.3	2.41	71.5	0.42	0.17	0.79
	Can	97.5	1.64	37.8	1.29	72.1	1.17	0.08	1.77
	Mug	97.6	1.65	21.3	1.46	62.4	0.40	0.11	1.25
	Laptop	90.5	1.88	20.1	1.00	69.1	0.42	0.13	1.20

4.4 Comparison and Evaluation on Our Dataset

Finally, to validate the real-world performance of our method in dynamic environments where the camera captures moving objects, we evaluate its performance on our real-world dataset with trained model. We consider five main categories: *Bowl*, *Bottle*, *Can*, *Mug* and *Laptop*. Since these categories align with those in the REAL275 dataset, their ground-truth CAD models are also identical. Table. 13 reports the performance of our method along with an available strong baseline for multi-object tasks, namely CenterSnap [45]. As it could be seen, our method shows a good performance that achieves an average accuracy of 96.6%, 26.8% and 67.0% for *IoU*50, 5°5 cm and 10°5 cm. In comparison with CenterSnap, our method improves pose estimation performance by 8.1%, 13.1%, and 40.8%, respectively. Additionally, Our method achieves superior performance in 3D shape reconstruction evaluation, achieving a mAP of 0.11 for 3D shape based on the metric of CD. Qualitative results are illustrated in Fig. 9, providing further evidence of the effectiveness of our method in practical scenarios.

5 LIMITATIONS AND FUTURE WORK

Despite our method shows promising performance, it still has some limitations. For example, our model's reliance on shape priors restricts its zero-shot generalization ability to unseen instances. In the future, we will consider two aspects: First, we plan to enhance zero-shot inference

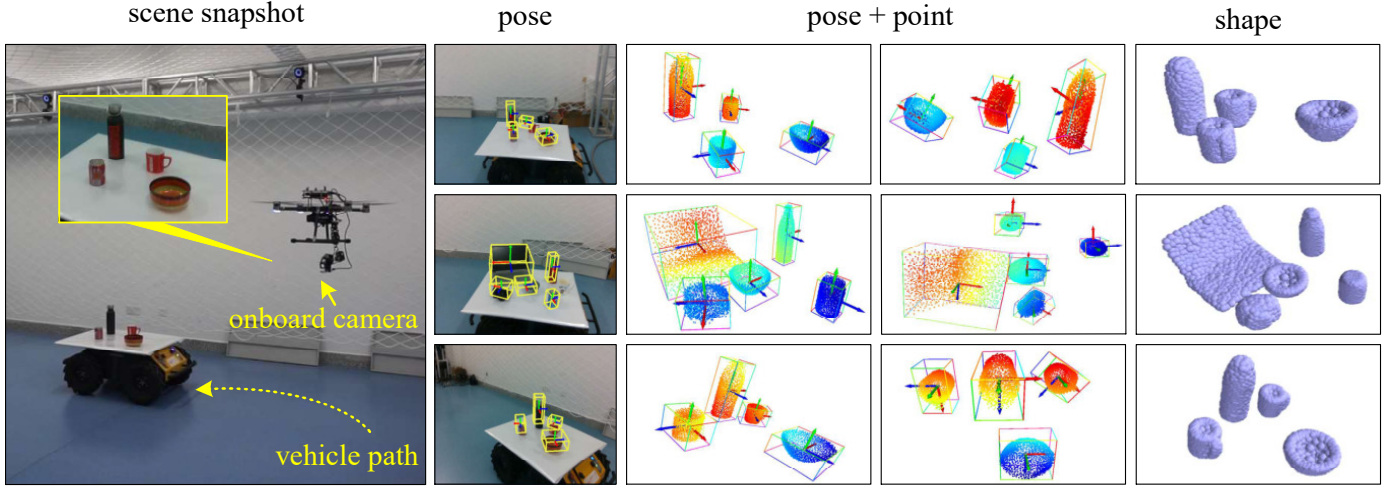


Fig. 9. Visualisation results of aerial dynamic scene on our dataset. Middle parts show the shape reconstruction from different viewpoints.

capabilities for pose tracking and shape reconstruction tasks. Second, inspired by the impressive capability of language description and large language models, we intend to investigate the potential of leveraging multimodal data to develop multimodal approaches for these tasks.

6 CONCLUSION

In this paper, we propose an effective self-supervised learning for multi-objective shape reconstruction and pose estimation, which only requires shape priors, eliminating the need for precisely labeling pose and CAD model for individual objects. Specifically, we introduce a diffusion-driven self-supervised network employing the Prior-Aware Pyramid 3D Point Transformer to explore the SE(3)-equivariant pose features and 3D scale-invariant shape information. Besides, we introduce a pretrain-to-refine self-supervised training paradigm that leverages priori shapes and shape/observation latent representation to train the proposed network. Extensive experiments conducted on four public datasets and our self-built test set show that our proposed method achieves superior performance in both self-supervised and fully-supervised pose estimation and shape reconstruction tasks.

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